Hyperuniformity of 1D Point Patterns: Project Summary

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March 2026 (updated after Week 3)

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1 Essential Theory

1.1 Hyperuniformity and its Classes

A point pattern is **hyperuniform** if its structure factor $S(k) \to 0$ as $k \to 0$. Equivalently, the number variance $\sigma^2(R)$ — variance of point counts in windows of half-width R — grows sub-linearly, beating the Poisson baseline $\sigma^2_{\text{Poisson}} = 2\rho R$.

Class	$S(k)$ **near** $k = 0$	**Variance**	α	**Metric**
I	$\sim B\|k\|^\alpha$, $\alpha > 1$	bounded	> 1	$\bar{\Lambda}$
II	$\sim B\|k\|$	$\sim C \ln R$	$= 1$	C_{II}
III	$\sim B\|k\|^\alpha$, $0 < \alpha < 1$	$\sim A \cdot R^{1-\alpha}$	$\in (0, 1)$	A_{III}
Stealthy	$= 0$ for all $\|k\| < K^*$	bounded	$\to \infty$	χ, K^*

1.2 Key Metrics

Surface-area coefficient $\bar{\Lambda}$ (Class I only):

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{1}{R} \int_0^R \sigma^2(R')\, dR'$$

This running average extracts the mean plateau level even when σ^2 oscillates. Lower $\bar{\Lambda}$ = less residual disorder. Minimum: $\bar{\Lambda} = 1/6 \approx 0.167$ (integer lattice, exact).

Generalized metrics for divergent classes:

$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}, \qquad A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}$$

1.3 Key Formulas

Eigenvalue formula for substitution tilings (Oğuz et al. 2019):

$$\boxed{\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}}$$

λ_1 = Perron-Frobenius eigenvalue of M; λ_2 = second-largest-magnitude eigenvalue. Hyperuniform iff $\alpha > 0$, i.e., $|\lambda_2| < \sqrt{|\lambda_1|}$.

2×2 determinant constraint: Any integer matrix with $|\det M| = 1$ has $|\lambda_1||\lambda_2| = 1$, so $|\lambda_2| = 1/|\lambda_1|$, giving $\alpha = 3$ always. This locks *every* metallic-mean chain to $\alpha = 3$ regardless of the specific mean.

URL (uncorrelated random lattice), displacement $\delta \sim \text{Uniform}[-a, a]$ (Klatt et al. 2020):

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6}, \qquad \alpha = 2 \text{ for all } a > 0.$$

Stealthy patterns, exclusion zone $K^* = 2\pi\rho\chi$ (formula derived for $d = 1$):

$$\bar{\Lambda} \approx \frac{1}{\pi^2 \chi}, \qquad \alpha \to \infty.$$

For stealthy patterns, α is not a meaningful power-law exponent; χ and K^* are the correct descriptors.

Stable distributions, stability index $s \in (0, 2]$: characteristic function $e^{-c^s|k|^s}$, so $\alpha = s$ exactly (continuously tunable).

Diffusion spreadability (Torquato 2021): at long times, $E(t) \sim t^{-(1+\alpha)/2}$ for Classes I/II/III. For quasicrystals, $E(t)$ oscillates log-periodically (period $2 \ln \lambda_1$ in $\log t$), making direct power-law fitting unreliable; use the eigenvalue formula instead.

2 Week 1: Benchmarks and the Metallic-Mean Family

2.1 What We Did

- Built and validated the core numerical pipeline: number variance estimator, structure factor, and spreadability code.
- Ran Poisson ($\sigma^2 = 2\rho R$ reproduced to 1.1%) and URL ($\bar{\Lambda} = (1 + a^2)/6$ reproduced to $< 0.3\%$) benchmarks.
- Characterized Fibonacci, Silver, and Bronze metallic-mean chains.
- Characterized URL-perturbed lattices with uniform, Gaussian, and stable-distribution displacements, spanning $\alpha \in (0, 2]$ continuously.
- Derived the $\alpha = 3$ determinant constraint algebraically.

2.2 Metallic-Mean Quasicrystals ($\alpha = 3$ for all)

Chain	$\bar{\Lambda}$	N
Fibonacci ($\lambda_1 \approx 1.618$)	0.201	14.9×10^6
Silver ($\lambda_1 \approx 2.414$)	0.250	22.6×10^6
Bronze ($\lambda_1 \approx 3.303$)	0.282	21.9×10^6

All three share $\alpha = 3$ due to the determinant constraint. The Fibonacci value 0.201 matches Zachary & Torquato (2009) analytical result to $< 0.5\%$.

2.3 Perturbed Lattices (URL and Stable Distributions)

Pattern	α	$\bar{\Lambda}$ (measured)	$\bar{\Lambda}$ (exact)
URL $a = 0.1$	2	0.168	0.168
URL $a = 0.5$	2	0.208	0.208
URL $a = 1.0$ (cloaked)	2	0.333	0.333
Stable $s = 1.7$	1.7	0.242	—
Stable $s = 1.5$	1.5	0.303	—
Stable $s = 1.3$	1.3	0.417	—
Cauchy ($s = 1$)	1	diverges	—
Stable $s = 0.5$	0.5	diverges	—

At $a = 1$ (cloaked URL), the sinc form factor vanishes at all Bragg positions; Bragg peaks disappear from $S(k)$ but hyperuniformity ($\alpha = 2$) is preserved.

2.4 Key Week 1 Finding

α and $\bar{\Lambda}$ are independent metrics. URL $a = 0.1$ ($\alpha = 2$, $\bar{\Lambda} = 0.168$) is nearly as ordered as the perfect lattice despite lower α than Fibonacci ($\alpha = 3$, $\bar{\Lambda} = 0.201$). The two metrics capture different aspects of order.

3 Week 2: Filling the Spectrum and the Full Catalog

3.1 What We Did

- Extended the metallic-mean family to Copper and Nickel (first reported $\bar{\Lambda}$ values).
- Characterized Period-Doubling (Class II, $\alpha = 1$ exact) and 0222 chain (Class III, $\alpha \approx 0.639$ exact).
- Identified the Bombieri-Taylor cubic substitution as the first known quasicrystal with $1 < \alpha < 2$ (specifically $\alpha = 1.545$).
- Incorporated stealthy ensemble data (from collaborator Sam, ∼4300 configs/χ).
- Assembled the full 27-pattern catalog.
- Wrote the Junior Paper, §1–§3 (Introduction, Theory, Methods).

3.2 Complete Metallic-Mean Family

Chain	$\bar{\Lambda}$	**Status**
Fibonacci ($n = 1$, $\lambda_1 \approx 1.618$)	0.201	Zachary & Torquato (2009)
Silver ($n = 2$, $\lambda_1 \approx 2.414$)	0.250	*novel*
Bronze ($n = 3$, $\lambda_1 \approx 3.303$)	0.282	*novel*
Copper ($n = 4$, $\lambda_1 \approx 4.236$)	0.293	*novel*
Nickel ($n = 5$, $\lambda_1 \approx 5.193$)	0.310	*novel*

$\bar{\Lambda}$ increases monotonically with metallic index n; increments diminish with n.

3.3 Bombieri-Taylor Cubic: Filling $1 < \alpha < 2$

Rule: $a \to aac$, $b \to ac$, $c \to b$. Matrix: $M_{\mathrm{BT}} = \left(\begin{smallmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. Characteristic polynomial: $x^3 - 2x^2 - x + 1 = 0$.

$$\lambda_1 = \theta_1 \approx 2.247, \quad |\lambda_2| \approx 0.802, \quad |\lambda_3| \approx 0.555.$$

Crucially, $|\lambda_2| = 0.802 \neq 1/\theta_1 = 0.445$: the 3×3 matrix breaks the unimodular constraint and yields:

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = 1.545, \qquad \bar{\Lambda} = 0.377.$$

Generated at $N \approx 9.5 \times 10^6$ (25 iterations). Confirmed Class I. This is the only known 1D quasicrystal with $1 < \alpha < 2$.

3.4 Class II and III Substitution Chains

Pattern	α	$\bar{\Lambda}$	**Class**
Period-Doubling ($M = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, $\lvert\lambda_2\rvert = 1$)	1 (exact)	diverges	II
0222 chain ($M = \left(\begin{smallmatrix} 0 & 2 \\ 2 & 2 \end{smallmatrix}\right)$, $\lvert\lambda_2\rvert = \sqrt{5} - 1 \approx 1.236$)	0.639 (exact)	diverges	III

3.5 Stealthy Patterns (collaborator data)

χ	Configs	$\bar{\Lambda}$ (measured)	$\bar{\Lambda}$ $(1/\pi^2\chi)$	Excess
0.1	4316	1.021	1.013	+0.8%
0.2	4316	0.526	0.507	+3.7%
0.3	4316	0.357	0.338	+5.6%

Small systematic excess from $S(k)$ overshoot just above K^*.

4 Week 3: Deeper Analysis and New Results

4.1 What We Did

Six investigations following the meeting with Professor Torquato:

1. BT deep dive: diagnostic of oscillations and spreadability failure
2. Finite-size scaling: convergence of α and $\bar{\Lambda}$ with N
3. New cubic tilings: systematic search; proved $2 < \alpha < 3$ inaccessible at degree 3
4. Generalized metrics: defined and measured C_{II} and A_{III}
5. $\bar{\Lambda}$ finite-N corrections: lattice analytic proof; stealthy two phases
6. Catalog cleanup: stealthy α=none; density invariance confirmed

4.2 Investigation 1: Bombieri-Taylor Diagnostic

BT has an unusually dense Bragg spectrum: **32,517 peaks in** $k < 50$ versus 2,637 for Fibonacci (12× denser). This makes spreadability unreliable:

- Rod overlap: minimum inter-point spacing ≈ 1.000, but two-phase rod diameter $2a = 1.002$ causes overlap; corrected to $a_{\text{rod}} = 0.45$.
- Spreadability at $t \in [10^2, 10^5]$: $\hat{\alpha} = 1.95$–2.02 (far above theory 1.545).
- Even at $t \in [10^5, 10^9]$: $\hat{\alpha} = 1.673$ (still converging slowly from above).
- Root cause: large $|\lambda_2| = 0.802$ produces strong log-periodic oscillations (period $2 \ln \theta_1 \approx 1.62$ in $\log t$) that defeat power-law fitting.

Conclusion: For BT, use $\bar{\Lambda} = 0.377$ from number variance, not spreadability. Spreadability would require $t \gg 10^9$ to converge.

4.3 Investigation 2: Finite-Size Scaling of Metrics

Direct $S(k)$ power-law fits fail for substitution tilings (Bragg-peak dominated). Using variance-based metrics:

- BT $\bar{\Lambda}$: stable at 0.378 from $N = 2{,}000$ (!). Very fast convergence.
- Fibonacci $\bar{\Lambda}$: stable at 0.199 from $N = 2{,}000$.
- Period-Doubling C_{II}: converges to 0.123 only for $N \gtrsim 100{,}000$.
- 0222 $\hat{\alpha}$: oscillates between 0.66–1.05 at $N = 1.5 \times 10^6$, does not converge to 0.639. Root cause: $|\lambda_2| = 1.236 > 1 \to$ diffuse spectrum $\to$ slow convergence.

4.4 Investigation 3: The $2 < \alpha < 3$ Gap is Inaccessible at Degree 3

Exhaustive search: All 3×3 non-negative integer matrices with row sums ≤ 4: 39,304 candidate matrices $\to$ 4,882 Pisot matrices $\to$ 305 apparently in $(2, 3)$.

Upon full floating-point evaluation, all 305 give $\alpha = 2.0000$ or $\alpha = 3.0000$ exactly.

Algebraic proof: For any 3×3 matrix with $|\det M| = 1$ and a complex conjugate eigenvalue pair $\lambda_2 = \bar{\lambda}_3$:

$$\lambda_1 |\lambda_2|^2 = |\det M| = 1 \implies |\lambda_2| = \frac{1}{\sqrt{\lambda_1}} \implies \alpha = 1 - \frac{-\ln \lambda_1}{\ln \lambda_1} = 2.$$

All-real eigenvalue cases with row sums ≤ 4 produce no $\alpha \in (2, 3)$ numerically.

Conclusion: The $2 < \alpha < 3$ gap requires algebraic degree ≥ 4 (4×4 matrices or larger entries). This is a proved impossibility at degree 3 within the row-sum ≤ 4 constraint.

Byproduct: the cubic α=2 tiling (new result). The matrix $M_{\alpha 2} = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{smallmatrix}\right)$, rule $a \to bc$, $b \to c$, $c \to abc$, has $\lambda_1 \approx 2.148$ and $|\lambda_2| \approx 0.682$. By the proof above, $\alpha = 2$ exactly. Simulation at $N \approx 10^6$ confirms Class I with $\bar{\Lambda} = 0.275$. To our knowledge, this is the **first deterministic quasiperiodic substitution tiling with** $\alpha = 2$ — a structural analogue of the stochastic URL.

4.5 Investigation 4: Generalized Metrics for Class II and III

Pattern	Metric	Value	How extracted
Period-Doubling	$C_{II} = \lim \sigma^2(R)/\ln R$	0.123 ± 0.016	plateau of $\sigma^2/\ln R$ at large R
0222 chain	$A_{III} = \lim \sigma^2(R)/R^{1-\alpha}$	0.151 ± 0.014	plateau using $\alpha = 0.639$ (exact)

`compute_generalized_metric()` added to `quasicrystal_variance.py`.

4.6 Investigation 5: $\bar{\Lambda}$ Finite-N Corrections and the Two Stealthy Phases

Integer lattice: $\bar{\Lambda}_{\text{analytic}}(R_{\max}) = 1/6$ exactly for all integer $R_{\max}$ (the running-average integral of $\sigma^2(R) = \{R\} - \{R\}^2$ is piecewise cubic and evaluates to exactly 1/6). Simulation confirms: all estimates within 0.3% of 1/6.

Stealthy two phases (key insight): Running `generate_stealthy()` from a perturbed-lattice initial condition and running the collaborator ensemble from random initial conditions gives radically different results:

Phase	Initial condition	$\bar{\Lambda}$
Ordered-stealthy	Near-lattice start	$\approx 1/6 = 0.167$
Disordered-stealthy	Random start (collab ensemble)	$\approx 1/(\pi^2\chi)$

Both satisfy $S(k) = 0$ for $|k| < K^*$ — both are valid stealthy configurations — but they differ dramatically in $\bar{\Lambda}$. Stealthy hyperuniformity does *not* uniquely determine a pattern; there is an order-disorder transition within the stealthy phase.

4.7 Investigation 6: Catalog Cleanup

- Stealthy α replaced with "—" (not a power law) in all tables and figures. $K^* = 2\pi\rho\chi$ added as the correct descriptor.
- **Density invariance:** $\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$ (numerically confirmed to $< 0.03\%$). Direct comparison of $\bar{\Lambda}$ values across patterns with different intrinsic densities is already valid.

4.8 Complete Updated Catalog

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	0.167
URL $a = 0.1$	I	2.0	0.168
Fibonacci	I	3.0	0.201
URL $a = 0.5$	I	2.0	0.208
Silver	I	3.0	0.250
Cubic α=2 (new)	*I*	*2.0*	*0.275*
Bronze	I	3.0	0.282
Copper	I	3.0	0.293
Nickel	I	3.0	0.310
URL $a = 1.0$ (cloaked)	I	2.0	0.333
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	**I**	**1.545**	**0.377**
Stealthy $\chi = 0.2$	I	—	0.526
Stealthy $\chi = 0.1$	I	—	1.021
Period-Doubling	II	1.0	div. ($C_{II} = 0.123$)
0222 Chain	III	0.64	div. ($A_{III} = 0.151$)

Table 1: Class I (green) ranked by $\bar{\Lambda}$. Class II (yellow) and III (pink) have divergent $\bar{\Lambda}$; generalized metrics shown. Stable distributions, Gaussian perturbations, and additional URL values omitted for brevity; see the full ranking table from Week 1/2.

5 Open Questions and Next Steps

These questions are the natural next directions for the project. Each subsection is written with enough context for an agent to investigate independently.

5.1 Are the Fibonacci and Silver $\bar{\Lambda}$ values exact?

Current state:

- Fibonacci: $\bar{\Lambda} = 0.201$ from Zachary & Torquato (2009), confirmed numerically to $< 0.5\%$ at $N = 14.9 \times 10^6$. The paper gives an analytical theta-function expression and quotes 0.20110. It is not obvious whether this is a "nice" exact number (like 1/5).
- Silver: $\bar{\Lambda} = 0.250 = 1/4$ from our Monte Carlo (± 0.0002). The number 1/4 is suspiciously round. No analytical derivation exists in the literature to our knowledge.

What to investigate:

1. Read Zachary & Torquato (2009), Table 1 carefully. What is their exact expression for the Fibonacci $\bar{\Lambda}$? Is it written as a theta-function sum that could evaluate to a simple fraction? Check whether $0.20110\ldots$ matches $1/5 = 0.200\ldots$ or a more exotic closed form.
2. For the Silver chain, the substitution matrix is $M = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ with $\lambda_1 = 1 + \sqrt{2}$ (silver ratio). The analytical framework in Zachary & Torquato generalizes to any metallic-mean chain. Try evaluating their formula with the Silver chain's eigenvalues and tile lengths; see if the result simplifies to 1/4.

3. If Silver is 1/4 exactly, check whether Bronze ($\bar{\Lambda} = 0.282$) is $\sqrt{2}/5$, and whether there is a pattern involving the metallic mean μ_n.

5.2 What does the metallic-mean $\bar{\Lambda}$ sequence converge to?

Current state: The sequence is $0.201, 0.250, 0.282, 0.293, 0.310, \ldots$ for $n = 1, 2, 3, 4, 5$. Increments: $+0.049, +0.032, +0.011, +0.017, \ldots$ (non-monotone; Copper/Nickel spacing may have numerical noise). $\bar{\Lambda}$ increases monotonically with n, so the sequence converges to some limit $\bar{\Lambda}_\infty \leq$ something.

A natural candidate for the limit: as $n \to \infty$, the metallic mean $\mu_n \approx n$ and the Silver ratio $\to$ large values. The URL at $a = 1$ (cloaked) gives $\bar{\Lambda} = 1/3$, and the cloaked lattice can be thought of as the limiting case where perturbations are maximal. It is plausible that $\bar{\Lambda}_\infty = 1/3$.

What to investigate:

1. Compute $\bar{\Lambda}$ for higher metallic-mean chains ($n = 6, 7, 8, 10, 20$) and plot $\bar{\Lambda}$ vs. $1/n$ to extrapolate the limit.

2. Try to derive the limit analytically. For large n, the substitution rule is $S \to L$, $L \to L^n S$ (a long string of L's). In this limit the tile-length ratio $\ell_L/\ell_S = \mu_n \approx n$, the density $\rho = 1/(1 + 1/\mu_n) \to 1$, and the pair-correlation structure approaches that of a specific limit. Does $\bar{\Lambda} \to 1/3$ exactly? (The URL cloaked value.) Or something else?

3. Compare the extrapolated limit to the Zachary & Torquato (2009) expression; see if taking $\lambda_1 \to \infty$ in their theta-function formula yields a closed form.

5.3 Can a higher-rank substitution matrix achieve $2 < \alpha < 3$?

Current state: Week 3 proved that all 3×3 unimodular Pisot matrices with a complex conjugate eigenvalue pair give $\alpha = 2$ exactly. An exhaustive search over 39,304 matrices (row sums ≤ 4) confirms no 3×3 case falls in $(2, 3)$.

The algebraic reason: any degree-3 polynomial with two complex conjugate roots $\lambda_2 = \bar{\lambda}_3$ and $|\det| = 1$ forces $|\lambda_2| = 1/\sqrt{\lambda_1}$, giving $\alpha = 2$. To get $\alpha \in (2, 3)$ one needs $|\lambda_2| \in (1/\lambda_1, 1/\sqrt{\lambda_1})$, which requires breaking the unimodular constraint, using all-real eigenvalues, or going to degree 4.

What to investigate:

1. 4×4 **matrices (highest priority):** Enumerate 4×4 non-negative integer matrices with row sums ≤ 3 (already a large search: $\sim 3^{16} = 43 \times 10^6$ candidates, but most filtered quickly by Pisot condition). Filter for $2 < \alpha < 3$ using the eigenvalue formula. The key is that for degree-4 characteristic polynomials, complex conjugate pairs do not exhaust all cases; there can be two pairs or mixed real/complex configurations that avoid the $\alpha = 2$ lock.

2. **Relaxed row-sum constraint for** 3×3**:** Our search only covered row sums ≤ 4. Try row sums ≤ 6 or ≤ 8 for 3×3 matrices. The degree-3 proof holds for unimodular determinants, but non-unimodular 3×3 Pisot matrices (possible if row sums are large) might escape the $\alpha = 2$ lock. Check: does $|\det M| \neq 1$ for large-entry 3×3 matrices, and if so, can this give $\alpha \in (2, 3)$?

3. The search script is `search_cubic_tilings.py`. Extend it to 4×4. The key functions to reuse: `is_pisot()`, eigenvalue formula, matrix enumeration. For 4×4, use the eigenvalue formula with the two largest-magnitude eigenvalues (λ_1 and λ_2) as before.

5.4 Is there an analytical formula for $\bar{\Lambda}$ of any metallic-mean chain?

Current state: The general method in Zachary & Torquato (2009) uses a theta-function representation of the pair-correlation function of a substitution tiling, valid for any PV substitution. They compute Fibonacci explicitly. In principle, the same method applied to Silver would confirm or refute $\bar{\Lambda}_{\text{Silver}} = 1/4$.

What to investigate: This is a more theoretical direction. Read Sections III–IV of Zachary & Torquato (2009) to understand their theta-function method. Try to apply it to the Silver chain ($\lambda_1 = 1+\sqrt{2}$, tile ratio $L/S = 1+\sqrt{2}$, frequencies $f_L = 1-1/\sqrt{2}$, $f_S = 1/\sqrt{2}-1/2$). The integral for $\bar{\Lambda}$ should reduce to a sum over Bragg-peak wavevectors with known intensities; evaluate numerically and compare to the Monte Carlo value 0.250.

6 Software and File Guide

File	Purpose
`substitution_tilings.py`	Chain generation: all 9 CHAINS including cubic_alpha2
`quasicrystal_variance.py`	Number variance + `compute_generalized_metric()`
`two_phase_media.py`	Spectral density + spreadability
`disordered_patterns.py`	URL, stealthy generation + `lambda_bar_url_exact()`
`research_catalog.py`	Full pipeline → `results/data/catalog.json`
`make_figures.py`	All figures from catalog.json
`analyze_bombieri_taylor.py`	BT analysis
`stealthy_analysis.py`	Collab stealthy ensemble (∼4316 configs/χ)
`bt_deep_dive.py`	Week 3 BT diagnostic (Inv 1)
`finite_size_scaling.py`	Week 3 metric convergence (Inv 2)
`search_cubic_tilings.py`	Week 3 3×3 search (Inv 3)
`generalized_metrics.py`	Week 3 C_{II}, A_{III} (Inv 4)
`finite_size_lambda_bar.py`	Week 3 finite-N $\bar{\Lambda}$ (Inv 5)
`catalog_cleanup.py`	Week 3 cleanup → `results/catalog_updated.json`
`jp/jp_hyperuniformity.tex`	Junior Paper (§1–§3 only; results TBD)
`jp/jp_results_draft.tex`	Drafted §4–§8, saved for later
`results/week3/week3_presentation.tex`	Week 3 slides
`KNOWLEDGE.md`	Technical reference (all definitions, formulas)
`CONTEXT.md`	Full project roadmap and phase log

7 Key References

1. Torquato & Stillinger (2003). *Phys. Rev. E* **68**, 041113. Foundational paper: hyperuniformity definition, $\bar{\Lambda}$, $\bar{\Lambda} = 1/6$ for lattice.

2. Torquato (2018). *Physics Reports* **745**, 1–95. Comprehensive review. Good for context and applications.

3. Oğuz, Socolar, Steinhardt & Torquato (2019). *Acta Cryst. A* **75**, 3. **Central theory reference.** Eigenvalue formula for α; classification of Period-Doubling and 0222 chains.

4. Bombieri & Taylor (1986). *J. Physique Colloques* **47**, C3-19. Source of the cubic substitution ($\alpha = 1.545$); characterizes which 1D tilings give pure Bragg diffraction.

5. Klatt, Kim & Torquato (2020). *Phys. Rev. E* **101**, 032118. Exact URL formula $\bar{\Lambda} = (1+a^2)/6$; cloaking effect at $a = 1$.

6. Torquato, Zhang & Stillinger (2015). *Phys. Rev. X* **5**, 021020. Stealthy ground states; ensemble theory.

7. Zachary & Torquato (2009). *J. Stat. Mech.* P12015. $\bar{\Lambda}$ analytical result for Fibonacci (0.20110); theta-function method applicable to other metallic-mean chains.

8. Torquato (2021). *Phys. Rev. E* **104**, 054102. Diffusion spreadability: $E(t) \sim t^{-(1+\alpha)/2}$.

9. Wang & Torquato (2022). *Phys. Rev. Applied* **17**, 034022. Spreadability as a dynamic measure of hyperuniformity.

10. Hitin-Bialus, Maher, Steinhardt & Torquato (2024). *Phys. Rev. E* **109**, 064108. Spreadability applied to quasiperiodic tilings; packing fraction $\phi_2 = 0.35$.